Inkspection: Detecting Document Forgery by Cross-Checking Pixel, Typographic and Semantic Evidence at the Word Level

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***Abstract*—Receipts and invoices are now routinely scanned, read by OCR and paid without anyone looking closely at the page. That makes a small edit, such as a changed total or date, surprisingly easy to get through. Current detectors tend to look at only one thing. Pixel-forensic methods can find where an image was touched, but they cannot say whether the change makes the document wrong. Rule-based checks catch totals that don't add up, yet they miss careful edits that keep the arithmetic intact, and they often mistake an OCR misread for fraud. We present Inkspection, a system of cooperating agents that collect pixel, typographic and semantic evidence and tie all of it to the same OCR words. A calibrated word-level model then scores the document, marks the altered regions and produces an audit report that points to the evidence behind each flag. On the Find It Again! receipt forgery benchmark, Inkspection reaches a document-level PR-AUC of [•], compared with [•] for the strongest single-modality baseline, and a word-level localisation F1 of [•]. At a 95% recall target, it sends [•]% of documents to human review instead of [•]%. Our results indicate that comparing evidence word by word, instead of averaging page-level scores, makes detection both more accurate and easier to audit.**

Keywords*—*document forgery detection, image forensics, optical character recognition, multimodal fusion, multi-agent systems, explainable AI

I. Introduction

Most companies no longer handle paper receipts by hand. Expense claims and supplier invoices arrive as scans or phone photos, OCR software reads them, and a payment system takes it from there. Often nobody looks at the image itself. For a fraudster this is convenient: a few minutes in an ordinary image editor is enough to raise an amount, move a date or swap a vendor's name.

These edits are hard to catch because they are so small. People forging a receipt rarely redo the whole page. They change a few characters and try to make them blend in. The Find It Again! dataset [1], built from receipts in the SROIE benchmark [4], is a good example. Its forgeries were made by volunteers using everyday tools such as Paint and GIMP, and in 353 of its 455 modified areas the forger simply copied characters from another part of the same receipt. An edit like this can leave two kinds of trace. Some are visual: a patch that compresses differently, noise that doesn't match, a digit drawn a little thinner than its neighbours. Others are logical, such as line items that no longer add up to the total.

Most existing detectors look for only one of these. Pixel-forensic methods, from error level analysis to learned tampered-text detectors [2], are good at finding where pixels were changed. What they cannot do is tell whether a change matters, and they are easily distracted by stamps, folds and pen marks that are perfectly innocent. Rule-based validators take the opposite approach. They work on the text that OCR systems such as PP-OCR [5] or TrOCR [3] produce, checking that items, tax and totals agree and that dates make sense. A careful edit that keeps the sums consistent slips straight past them. And because a misread digit breaks a sum just as an edited one does, they raise false alarms whenever the OCR makes a mistake. Used alone, each approach leaves an opening that a careful forger can exploit.

Our view is that the two kinds of evidence should be checked against each other, and that the comparison has to happen word by word. Take an amount that breaks the receipt's arithmetic. If that same amount is also printed slightly thinner, sits a pixel or two off the baseline and recompresses differently from the words around it, something is almost certainly wrong. If the only clue is a failed sum on a word the OCR engine was unsure how to read, it may be nothing at all. One pixel score and one text score for the whole page cannot tell these cases apart.

Inkspection is built around this idea, with the work split between several agents. The orchestrator keeps the original image untouched for forensic analysis and makes a separate, straightened copy for OCR. The vision agent runs six pixel-forensic detectors on the original. At the same time, the OCR and layout agent reads the words and compares each one's typography with the rest of its line. The semantic agent then pulls out the receipt's fields and checks quantities, sums, tax, change and dates, noting exactly which words each failed check involves. Last, the adjudicator combines everything word by word using a gradient-boosted model. It calibrates the result and returns a document score, a review tier, the suspicious regions and an audit report that cites the evidence for each finding.

To test whether this helps, we compare Inkspection with four traditional detectors: a classic error level analysis threshold, a vision-only classifier, a rules-only validator, and a naive fusion that averages the last two. All five systems are scored from the same cached agent outputs, so any difference comes from how the evidence is combined rather than from different preprocessing. Along with the usual detection and localisation

metrics, we report how many documents each system would send to a human reviewer at 95% recall, since that is what auditing actually costs.

Our main contributions are:

1) A multi-agent architecture that collects pixel-forensic, typographic and semantic evidence in parallel and links all of it to the same OCR words, while forensic analysis always runs on the unmodified image.
2) A word-level fusion model with explicit cross-modal agreement features and calibrated scores, which localises the altered regions and explains each one through the evidence behind it.
3) An evaluation on Find It Again! that compares Inkspection with single-modality and naive-fusion baselines on identical inputs, reporting review workload at fixed recall and catch rates at an equal false-alarm rate.
4) An ablation study of what each source of evidence contributes, together with an open-source implementation that includes a web interface for analysis and human review.

The rest of the paper is organised as follows. Section II covers related work on document forgery detection and OCR. Section III describes the agents and the fusion model, and Section IV the dataset, baselines and metrics. Section V presents the results, and Section VI discusses limitations and future work.

REFERENCES

[1] B. Martínez Tornés, T. Taburet, E. Boros, K. Rouis, A. Doucet, P. Gomez-Krämer, N. Sidere, and V. Poulain d'Andecy, "Receipt dataset for document forgery detection," in *Proc. 17th Int. Conf. Document Analysis and Recognition (ICDAR)*, 2023, pp. 454–469, doi: 10.1007/978-3-031-41682-8_28.

[2] C. Qu, C. Liu, Y. Liu, X. Chen, D. Peng, F. Guo, and L. Jin, "Towards robust tampered text detection in document image: New dataset and new solution," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 5937–5946.

[3] M. Li, T. Lv, J. Chen, L. Cui, Y. Lu, D. Florencio, C. Zhang, Z. Li, and F. Wei, "TrOCR: Transformer-based optical character recognition with pre-trained models," in *Proc. AAAI Conf. Artificial Intelligence*, vol. 37, no. 11, 2023, pp. 13094–13102.

[4] Z. Huang, K. Chen, J. He, X. Bai, D. Karatzas, S. Lu, and C. V. Jawahar, "ICDAR2019 competition on scanned receipt OCR and information extraction," in *Proc. Int. Conf. Document Analysis and Recognition (ICDAR)*, 2019, doi: 10.1109/ICDAR.2019.00244.

[5] Y. Du, C. Li, R. Guo, X. Yin, W. Liu, J. Zhou, Y. Bai, Z. Yu, Y. Yang, Q. Dang, and H. Wang, "PP-OCR: A practical ultra lightweight OCR system," arXiv:2009.09941, 2020.